

Assignment 2- CP363

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Group # 26

Equipment Test Center

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MySQL Source Code & Table Definitions

```
1 • CREATE DATABASE IF NOT EXISTS EquipmentTestCenterDB;
2 • USE EquipmentTestCenterDB;
3
4 • CREATE TABLE Equipment (
5     Equipment_ID INT PRIMARY KEY,
6     Name VARCHAR(100) NOT NULL,
7     Model_Number VARCHAR(50) NOT NULL,
8     Manufacturer VARCHAR(100) NOT NULL,
9     Status VARCHAR(20) NOT NULL CHECK (Status IN ('Available','In Use','Under Maintenance'))
10 );
11
12 • CREATE TABLE Client (
13     Client_ID INT PRIMARY KEY,
14     First_Name VARCHAR(50) NOT NULL,
15     Last_Name VARCHAR(50) NOT NULL,
16     Contact_Email VARCHAR(100) NOT NULL UNIQUE,
17     Phone_Number VARCHAR(20) NOT NULL UNIQUE,
18     Organization VARCHAR(100) NOT NULL
19 );
20
21 • CREATE TABLE Technician (
22     Technician_ID INT PRIMARY KEY,
23     First_Name VARCHAR(50) NOT NULL,
24     Last_Name VARCHAR(50) NOT NULL,
25     Certification VARCHAR(100) NOT NULL,
26     Contact_Email VARCHAR(100) NOT NULL UNIQUE,
27     Phone_Number VARCHAR(20) NOT NULL UNIQUE
28 );
29
30 • CREATE TABLE Test (
31     Test_ID INT PRIMARY KEY,
32     Test_Name VARCHAR(100) NOT NULL,
33     Description VARCHAR(255) NOT NULL
34 );
35
36 • CREATE TABLE Lab (
37     Lab_ID INT PRIMARY KEY,
38     Lab_Name VARCHAR(100) NOT NULL,
39     Location VARCHAR(100) NOT NULL,
40     Capacity INT NOT NULL CHECK (Capacity > 0)
41 );
42
43 • CREATE TABLE Booking (
44     Booking_ID INT PRIMARY KEY,
45     Booking_Date DATE NOT NULL,
46     Client_ID INT NOT NULL,
47     Lab_ID INT NOT NULL,
48     Start_Time TIME NOT NULL,
49     End_Time TIME NOT NULL,
50     Status VARCHAR(20) NOT NULL CHECK (Status IN ('Pending','Confirmed','Completed')
51     FOREIGN KEY (Client_ID) REFERENCES Client(Client_ID),
52     FOREIGN KEY (Lab_ID) REFERENCES Lab(Lab_ID)
53 );
```

```

54
55 • CREATE TABLE Test_Session (
56     Session_ID INT PRIMARY KEY,
57     Test_Date DATE NOT NULL,
58     Result VARCHAR(100) NOT NULL,
59     Duration TIME NOT NULL,
60     Technician_ID INT NOT NULL,
61     Test_ID INT NOT NULL,
62     FOREIGN KEY (Technician_ID) REFERENCES Technician(Technician_ID),
63     FOREIGN KEY (Test_ID) REFERENCES Test(Test_ID)
64 );
65
66 • CREATE TABLE Report (
67     Report_ID INT PRIMARY KEY,
68     Report_Date DATE NOT NULL,
69     Findings VARCHAR(255) NOT NULL,
70     Approved_By INT NOT NULL,
71     Session_ID INT NOT NULL UNIQUE,
72     FOREIGN KEY (Approved_By) REFERENCES Technician(Technician_ID),
73     FOREIGN KEY (Session_ID) REFERENCES Test_Session(Session_ID)
74 );
75
76 • CREATE TABLE Invoice (
77     Booking_ID INT NOT NULL,
78     Invoice_No INT NOT NULL,
79     Invoice_Date DATE NOT NULL,
80     Amount DECIMAL(10,2) NOT NULL CHECK (Amount >= 0),
81     Payment_Status ENUM('Paid','Unpaid') NOT NULL,
82     PRIMARY KEY (Booking_ID, Invoice_No),
83     FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID)
84 );
85
86 • CREATE TABLE Maintenance_Record (
87     Equipment_ID INT NOT NULL,
88     Record_No INT NOT NULL,
89     Maintenance_Date DATE NOT NULL,
90     Technician_ID INT NOT NULL,
91     Notes VARCHAR(255) NOT NULL,
92     PRIMARY KEY (Equipment_ID, Record_No),
93     FOREIGN KEY (Equipment_ID) REFERENCES Equipment(Equipment_ID),
94     FOREIGN KEY (Technician_ID) REFERENCES Technician(Technician_ID)
95 );
96

```

```

97      -- Junction table for Booking <-> Equipment (M:N)
98  ● ○ CREATE TABLE Booking_Equipment (
99      Booking_ID INT NOT NULL,
100     Equipment_ID INT NOT NULL,
101     PRIMARY KEY (Booking_ID, Equipment_ID),
102     FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID),
103     FOREIGN KEY (Equipment_ID) REFERENCES Equipment(Equipment_ID)
104  );
105
106      -- Junction table for Test_Session <-> Equipment (M:N)
107  ● ○ CREATE TABLE TestSession_Equipment (
108     Session_ID INT NOT NULL,
109     Equipment_ID INT NOT NULL,
110     PRIMARY KEY (Session_ID, Equipment_ID),
111     FOREIGN KEY (Session_ID) REFERENCES Test_Session(Session_ID),
112     FOREIGN KEY (Equipment_ID) REFERENCES Equipment(Equipment_ID)
113  );

```

Referential Integrity Explanation

A "Primary Key" (PK), such as Equipment_ID in Equipment and Client_ID in Client, uniquely identifies each record in the Equipment Test Center database and avoids duplicates. By guaranteeing the presence of referenced data, "Foreign Keys" (FK), connect tables and maintain referential integrity. Take Booking, for example. Client is denoted by Client_ID. Client_ID, which stops a reservation for an invalid customer. To guarantee that every test session has a legitimate technician, Test_Session.Technician_ID also makes reference to Technician.Technician_ID. By maintaining many-to-many relationships with composite PKs and FKs, junction tables such as Booking_Equipment make sure that every link points to legitimate entries. Together, these limitations protect the database relationships' dependability and consistency.

Relational Schema Diagram

